Fractions, Decimals and Percentages

Name _____ Date _____ No calculator Version 1 Scan to practise this topic online

KEY WORDS fraction *(hautanga)* · decimal *(tau ā-ira)* · percentage *(ōrau)* · equivalent *(ōrite)*

A Tenths and hundredths ●○○

⚠ **Watch out:** The first decimal place is tenths and the second is hundredths: $\frac{7}{10} = 0.7$ but $\frac{7}{100} = 0.07$.

1. Write each fraction as a decimal.

a) $\frac{3}{10}$ = _____ **b)** $\frac{7}{10}$ = _____ **c)** $\frac{9}{100}$ = _____ **d)** $\frac{23}{100}$ = _____

e) $\frac{1}{10}$ = _____ **f)** $\frac{61}{100}$ = _____ **g)** $\frac{1}{100}$ = _____ **h)** $\frac{99}{100}$ = _____

2. Write each decimal as a fraction in its simplest form.

a) 0.1 = _____ **b)** 0.7 = _____ **c)** 0.03 = _____ **d)** 0.47 = _____

e) 0.5 = _____ **f)** 0.2 = _____ **g)** 0.25 = _____ **h)** 0.75 = _____

B Percentages ●●○

💡 **Remember:** Per cent means "out of 100". 37% is 37 out of 100, which is $\frac{37}{100}$ or 0.37.

3. Write each percentage as a decimal.

a) 50% = _____ **b)** 25% = _____ **c)** 70% = _____ **d)** 8% = _____

e) 99% = _____ **f)** 10% = _____ **g)** 3% = _____ **h)** 45% = _____

4. Write each decimal as a percentage.

a) 0.45 = _____ **b)** 0.7 = _____ **c)** 0.09 = _____ **d)** 0.3 = _____

e) 0.01 = _____ **f)** 0.85 = _____ **g)** 0.5 = _____ **h)** 0.25 = _____

C Fractions and percentages ●●○

💡 **Remember:** Learn these: $\frac{1}{2} = 50\%$, $\frac{1}{4} = 25\%$, $\frac{3}{4} = 75\%$, $\frac{1}{10} = 10\%$ and $\frac{1}{5} = 20\%$.

5. Write each fraction as a percentage.

a) $\frac{1}{2}$ = _____ **b)** $\frac{1}{4}$ = _____ **c)** $\frac{1}{10}$ = _____ **d)** $\frac{3}{4}$ = _____

e) $\frac{1}{5}$ = _____ **f)** $\frac{3}{10}$ = _____ **g)** $\frac{2}{5}$ = _____ **h)** $\frac{9}{10}$ = _____

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6. Write each percentage as a fraction in its simplest form.

a) 50% = _____ **b)** 25% = _____ **c)** 10% = _____ **d)** 75% = _____

e) 20% = _____ **f)** 30% = _____ **g)** 1% = _____ **h)** 60% = _____

D All three together ●●○

7. Complete the table. Write each fraction in its simplest form.

Fraction	Decimal	Percentage
$\frac{1}{2}$		
	0.25	
		10%
$\frac{3}{4}$		
	0.2	
		70%

E Which is larger? ●●●

⚠ **Watch out:** Do not compare the digits you can see. Change both to percentages first.

8. Which is larger?

a) $\frac{1}{2}$ or 0.4 **b)** 30% or $\frac{1}{4}$ **c)** 0.7 or 65%

d) $\frac{3}{4}$ or 80% **e)** 0.09 or $\frac{1}{10}$ **f)** $\frac{1}{5}$ or 25%

F Problems ●●●

9. Aroha gets $\frac{7}{10}$ of a test right. Ben gets 65% of the same test right. Who did better? Show your working.

10. $\frac{1}{4}$ of the students in a class wear glasses. What percentage of the class do not wear glasses?

★ Extension

11. Write $\frac{13}{20}$ as a percentage. Start by changing it into hundredths.

12. A square is split into 100 small squares and 35 of them are shaded. What fraction of the square is shaded? Give your answer in its simplest form.

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Fractions, Decimals and Percentages

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS fraction *(hautanga)* · decimal *(tau ā-ira)* · percentage *(ōrau)* · equivalent *(ōrite)*

A Tenths and hundredths ●○○

⚠ **Watch out:** The first decimal place is tenths and the second is hundredths: $\frac{7}{10} = 0.7$ but $\frac{7}{100} = 0.07$.

1. Write each fraction as a decimal.

a) $\frac{1}{10}$ = _____ **b)** $\frac{9}{10}$ = _____ **c)** $\frac{7}{100}$ = _____ **d)** $\frac{37}{100}$ = _____

e) $\frac{3}{10}$ = _____ **f)** $\frac{53}{100}$ = _____ **g)** $\frac{3}{100}$ = _____ **h)** $\frac{91}{100}$ = _____

2. Write each decimal as a fraction in its simplest form.

a) 0.3 = _____ **b)** 0.9 = _____ **c)** 0.07 = _____ **d)** 0.59 = _____

e) 0.5 = _____ **f)** 0.4 = _____ **g)** 0.25 = _____ **h)** 0.75 = _____

B Percentages ●●○

💡 **Remember:** Per cent means "out of 100". 37% is 37 out of 100, which is $\frac{37}{100}$ or 0.37.

3. Write each percentage as a decimal.

a) 75% = _____ **b)** 40% = _____ **c)** 90% = _____ **d)** 6% = _____

e) 88% = _____ **f)** 20% = _____ **g)** 2% = _____ **h)** 35% = _____

4. Write each decimal as a percentage.

a) 0.35 = _____ **b)** 0.9 = _____ **c)** 0.04 = _____ **d)** 0.6 = _____

e) 0.07 = _____ **f)** 0.95 = _____ **g)** 0.5 = _____ **h)** 0.75 = _____

C Fractions and percentages ●●○

💡 **Remember:** Learn these: $\frac{1}{2} = 50\%$, $\frac{1}{4} = 25\%$, $\frac{3}{4} = 75\%$, $\frac{1}{10} = 10\%$ and $\frac{1}{5} = 20\%$.

5. Write each fraction as a percentage.

a) $\frac{1}{4}$ = _____ **b)** $\frac{1}{2}$ = _____ **c)** $\frac{1}{5}$ = _____ **d)** $\frac{1}{10}$ = _____

e) $\frac{3}{4}$ = _____ **f)** $\frac{7}{10}$ = _____ **g)** $\frac{3}{5}$ = _____ **h)** $\frac{4}{5}$ = _____

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6. Write each percentage as a fraction in its simplest form.

a) 25% = _____ **b)** 50% = _____ **c)** 20% = _____ **d)** 10% = _____

e) 75% = _____ **f)** 70% = _____ **g)** 1% = _____ **h)** 40% = _____

D All three together ●●○

7. Complete the table. Write each fraction in its simplest form.

Fraction	Decimal	Percentage
$\frac{1}{4}$		
	0.5	
		20%
$\frac{1}{10}$		
	0.75	
		30%

E Which is larger? ●●●

⚠ **Watch out:** Do not compare the digits you can see. Change both to percentages first.

8. Which is larger?

a) $\frac{1}{2}$ or 0.6 **b)** 20% or $\frac{1}{4}$ **c)** 0.8 or 75%

d) $\frac{3}{4}$ or 70% **e)** 0.08 or $\frac{1}{10}$ **f)** $\frac{1}{5}$ or 15%

F Problems ●●●

9. Aroha gets $\frac{4}{5}$ of a test right. Ben gets 85% of the same test right. Who did better? Show your working.

10. $\frac{1}{5}$ of the students in a class wear glasses. What percentage of the class do not wear glasses?

★ Extension

11. Write $\frac{9}{20}$ as a percentage. Start by changing it into hundredths.

12. A square is split into 100 small squares and 45 of them are shaded. What fraction of the square is shaded? Give your answer in its simplest form.

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